

- 1) A joint ensemble XY has the following joint distribution

$P(X,Y)$		X			
		1	2	3	4
Y	1	$1/8$	$1/16$	$1/32$	$1/32$
	2	$1/16$	$1/8$	$1/32$	$1/32$
	3	$1/16$	$1/16$	$1/16$	$1/16$
	4	$1/4$	0	0	0

calculate

- 1) what is the joint entropy $H(X,Y)$?

$$H(X,Y) = - \sum_x \sum_y P(x,y) \log_2 P(x,y)$$

using joint probabilities.

$P(X,Y)$	$Y=1$	$Y=2$	$Y=3$	$Y=4$
$X=1$	$1/8$	$1/16$	$1/32$	$1/32$
$X=2$	$1/16$	$1/8$	$1/32$	$1/32$
$X=3$	$1/16$	$1/16$	$1/16$	$1/16$
$X=4$	$1/4$	0	0	0

for each non-zero $P(x,y)$

$$\text{Contribution} = -P(x,y) \cdot \log_2(P(x,y))$$

$P(x,y)$	Contribution to $H(X,Y)$
$1/8$	$-(1/8) \cdot \log_2(1/8) = 0.375$
$1/16$	$-(1/16) \cdot \log_2(1/16) = 0.25$
$1/32$	$-(1/32) \cdot \log_2(1/32) = 0.15625$
$1/4$	$-(1/4) \cdot \log_2(1/4) = 0.5$

Summing all-

$$H(X,Y) = 3.375 \text{ bits} \checkmark$$

2) What are the marginal entropies $H(X)$ & $H(Y)$?

Soln $H(X)$:
First, $P(X)$ by summing over Y

X	$P(X)$
1	$1/8 + 1/16 + 1/32 + 1/32 = 1/4$
2	$1/16 + 1/8 + 1/32 + 1/32 = 1/4$
3	$1/16 + 1/16 + 1/16 + 1/16 = 1/4$
4	$1/4 + 0 + 0 = 1/4$

$$H(X) = -\sum_x P(X=x) \cdot \log_2 P(X=x)$$

$$H(X) = -(1/4 \cdot \log_2(1/4)) \times 4 = 2.0 \text{ bits/s}$$

$H(Y)$:

$P(Y)$ by summing over X

Y	$P(Y)$
1	$1/8 + 1/16 + 1/16 + 1/4 = 7/16$
2	$1/16 + 1/8 + 1/16 + 0 = 5/16$
3	$1/32 + 1/32 + 1/16 + 0 = 3/16$
4	$1/32 + 1/32 + 1/16 + 0 = 3/16$

$$H(Y) = -\sum_y P(Y=y) \cdot \log_2 P(Y=y)$$

$$= -(7/16 \cdot \log_2(7/16) + 5/16 \cdot \log_2(5/16) + 3/16 \cdot \log_2(3/16) + 3/16 \cdot \log_2(3/16))$$

$$= 2.75 \text{ bits/s}$$

3) For each value of y , what is the conditional entropy $H(X|y)$?

solⁿ $H(X|y) = - \sum_x P(X=x|Y=y) \cdot \log_2(P(X=x|Y=y))$

for $y=1$:

$$P(Y=1) = 7/16$$

$$P(X|Y=1) :$$

$$P(X=1|Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{1/8}{7/16} = 2/7$$

$$P(X=2|Y=1) = 1/7$$

$$P(X=3|Y=1) = 1/7$$

$$P(X=4|Y=1) = 4/7$$

$$H(X|Y=1) = -(2/7 \cdot \log_2(2/7) + 1/7 \cdot \log_2(1/7) + 1/7 \cdot \log_2(1/7) + 4/7 \cdot \log_2(4/7))$$

$$= -(0.2867 + 0.2102 + 0.2102 + 1.0438) = 1.75 \text{ bits}$$

$$H(X|Y=1) = 1.75 \text{ bits}$$

for $y=2$

$$P(Y=2) = 5/16$$

$$P(X|Y=2)$$

$$P(X=1|Y=2) = 1/5$$

$$P(X=2|Y=2) = 2/5$$

$$P(X=3|Y=2) = 1/5$$

$$P(X=4|Y=2) = 0$$

$$H(X|Y=2) = -(1/5 \cdot \log_2(1/5) + 2/5 \cdot \log_2(2/5) + 1/5 \cdot \log_2(1/5) + 0)$$

$$H(X|Y=2) = -(0.4644 + 0.5288 + 0.4644)$$

$$= 1.5 \text{ bits}$$

for $y=3$

$$P(Y=3) = 3/16$$

$$P(X|Y=3)$$

$$P(X=1|Y=3) = 1/6$$

$$P(X=2|Y=3) = 1/6$$

$$P(X=3|Y=3) = 1/3$$

$$P(X=4|Y=3) = 0$$

$$H(X|Y=3) = -(1/6 \cdot \log_2(1/6) + 1/6 \cdot \log_2(1/6) + 1/3 \cdot \log_2(1/3) + 0)$$

$$H(X|Y=3) = -(0.4308 + 0.4308 + 0.6309)$$

$$= 1.5 \text{ bits}$$

for $y=4$

$$P(Y=4) = 3/16$$

$$P(X|Y=4)$$

$$P(X=1|Y=4) = 1/6$$

$$P(X=2|Y=4) = 1/6$$

$$P(X=3|Y=4) = 1/3$$

$$P(X=4|Y=4) = 0$$

$$H(X|Y=4) = -(1/6 \cdot \log_2(1/6) + 1/6 \cdot \log_2(1/6) + 1/3 \cdot \log_2(1/3) + 0)$$

$$H(X|Y=4) = -(0.4308 + 0.4308 + 0.6309)$$

$$= 1.5 \text{ bits}$$

4) what is the conditional entropy $H(Y|X)$!

sol)

$$H(Y|X) = H(X, Y) - H(X)$$

$$H(Y|X) = 3.375 - 2.0 = 1.375 \text{ bits}$$

5) what is the conditional entropy of X given Y ? $H(X|Y)$

sol)

$$P(Y=1) = 7/16, H(X|Y=1) = 1.75$$

$$P(Y=2) = 5/16, H(X|Y=2) = 1.5$$

$$P(Y=3) = 3/16, H(X|Y=3) = 1.5$$

$$P(Y=4) = 1/16, H(X|Y=4) = 1.5$$

$$H(X|Y) = \sum_y P(Y=y) \cdot H(X|y)$$

$$= \left(\frac{7}{16} \times 1.75 \right) + \left(\frac{5}{16} \times 1.5 \right) + \left(\frac{3}{16} \times 1.5 \right) + \left(\frac{1}{16} \times 1.5 \right)$$

$$= 0.765625 + 0.46875 + 0.28125 + 0.09375$$

$$H(X|Y) = \underline{\underline{1.625 \text{ bits}}}$$

6) what is the mutual information between X and Y ?

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

$$I(X; Y) = 2.0 + 1.75 - 3.375$$

$$I(X; Y) = 0.375 \text{ bits}$$